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Sergio Orozco

https://sergiomorozco.github.io/ sergio_orozco@brown.edu

Research Interest

I'm interested in learning world models and the structure around them to support counterfacutal reasoning on long-horizon tasks. My research focuses on establishing which priors (learned or explicit) are useful for downstream robotic manipulation.

Education

Brown University — October 2023 - Current
Doctor of Philosophy in Computer Science — *GPA: 4.00*

University of Colorado at Colorado Springs — August 2016 - May 2019
Bachelor of Science in Computer Science — *GPA: 3.85*

Publications

Do Robots Need World Models? 2026 @ CORL 2026

TAP-MPC: Transferable Action Priors for Model Predictive Control. ***Sergio Orozco****, Chalyfey Vutha, Nandika Auluck, Anthony Ramirez-Roca, George Konidaris*

PhysWM 2026 @ CORL 2026

DefFlow: Goal-Conditioned Deformable Object Manipulation using 3D Point Flows. *Gaurav Singh,* ***Sergio Orozco****, Ryan Duong, Mete Tuluhan Akbulut, George Konidaris, Srinath Sridhar*

ReALM-GEN 2026 @ ICLR 2026

Adapting Diffusion Policies to Novel Environments via Policy-Steered Optimization. *Skye Thompson,* ***Sergio Orozco****, Eric Rosen, Karl Schmeckpeper, George Konidaris*

Best Paper Award
RINO @ CoRL 2025
In Submission
RA-L 2026

Learning Equivariant Neural-Augmented Object Dynamics from Few Interactions. ***Sergio Orozco****, Tushar Kusnur, Brandon May, Laura Herlant, George Konidaris.*

ICRA 2024

Robot Task Planning under Local Observability. *Max Merlin, Shane Parr, Neev Parikh,* ***Sergio Orozco****, Vedant Gupta, Eric Rosen, George Konidaris.*

CoRL 2023

Synthesizing Navigation Abstractions for Planning with Portable Manipulation Skills. *Eric Rosen, Steven James,* ***Sergio Orozco****, Vedant Gupta, Max Merlin, Stefanie Tellex, George Konidaris.*

Teaching

Teaching Assistant - Introduction to Neural Networks: Phil. of AI — July 2024 – August 2024
Brown University — *Providence, Rhode Island*

- Co-led daily discussions with 23 high school students on the impact of AI and its implications for society.
- Helped students write, execute, and debug various deep learning projects in Python and Tensorflow ranging from language translation to sequential decision making.

Graduate Student Mentor - NSF REU Site — June 2024 – August 2026
Brown University — *Providence, Rhode Island*

- Mentored 8 (2024) and 11(2026) undergraduate students from Minority Serving Institutions (MSIs) in The Leadership Alliance national consortium.
- Led weekly study groups aimed at teaching students "soft" research skills, necessary for any line of research.

Teaching Assistant - Introduction To Robotics September 2023 – December 2023
Brown University *Providence, Rhode Island*

- Helped 30 students develop and deploy robotic drones equipped with safety restrictions, networking capabilities, state estimators, controllers, and high-level planners.

Experience

Foundation Model Researcher June 2026 – Current
Midcentury *New York City, New York*

- Post-trained $\pi_{0.5}$ on proprietary egocentric and robot data to learn more robust policies.
- Led research project on a team of 3 people which resulted in a publication to a workshop at CoRL..
- Technologies Used: Python, Pytorch, Modal.

Research Intern February 2025 – August 2025
Robotics and AI Institute (RAI) *Cambridge, Massachusetts*

- Led research project on a team of 5 people which resulted in the **Best Paper Award** at CoRL RINO Workshop.
- Conducted experiments in simulation (IsaacSim, Genesis) and on real hardware (Franka Research Arm 3).
- Trained a world model to accurately learn the dynamics of 5 common household deformable objects with few interactions using equivariance and physics priors.
- Technologies Used: Python, Pytorch, Torch Geometric.

Research Assistant September 2022 – October 2023
Brown University *Providence, Rhode Island*

- Assisted and conducted published research in Humans to Robots Laboratory and Intelligent Robot Laboratory.
- Developed a testbed on a Boston Dynamics Spot robot which facilitated research on two projects.
- Equipped Spot to recognize ≈ 10 household items along with their corresponding locations and grasps which equipped researchers to test a multitude of novel task planners.
- Technologies Used: Python, BosdynSDK, Pytorch.

Flight Software Engineer June 2021 – September 2023
York Space Systems *Denver, Colorado*

- Designed and developed simulation software to emulate over 40 payload and critical bus components for York's S-Class satellites.
- Simulated over 500 commands and over 1,000 telemetry items processed by the onboard flight computer.
- Provided customers with the capabilities of York's 5 million dollar S-Class Satellite for testing without needing to provide any actual hardware.
- Designed and developed Store and Forward capabilities on York's S-Class satellite to ingest over 100 Jreap Messages per second using Google Protocol Buffers.
- Participated in several technical software interviews to search for potential talent amongst applicants.
- Technologies Used: C# .NET, Google Protocol Buffers, C, CMake, Make, Docker, Coverity.

Software Engineer June 2019 – May 2021
Sierra Nevada Corporation *Denver, Colorado*

- Developed and maintained a 23 million dollar simulation program that emulated realistic signals intelligence for the United States Air Force.
- Communicated and collaborated with customers to create realistic radio signals.
- Developed mission application systems for over 200 MH-60 helicopters capable of displaying IR video feed and Cursor on Target data to a world map.
- Lead and mentored college interns to develop proprietary tools for the company.
- Technologies Used: C# .NET, Reactive, Python, Windows Presentation Forms, Windows Form Application.